

Unit 9: Quadratics & Polynomials

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November 6, 2025

Session 9.1:

Exponents, Polynomial Operations, & GCF Factorin

Quick Reference: Exponent Rules

The Laws of Exponents

Product Rule:

$$x^a \cdot x^b = x^{a+b}$$

Quotient Rule:

$$\frac{x^a}{x^b} = x^{a-b} \text{ (when } x \neq 0\text{)}$$

Power Rule:

$$(x^a)^b = x^{ab}$$

Power of a Product:

$$(xy)^a = x^a y^a$$

Negative Exponent:

$$x^{-a} = \frac{1}{x^a} \text{ (when } x \neq 0\text{)}$$

Polynomial Operations

Multiplying: Use distributive property

Greatest Common Factor (GCF)

To factor using GCF:

1. Find the largest number that divides all coefficients
2. Find the lowest power of each variable in all terms
3. Factor out the GCF

Homework 9.1:

Exponents, Polynomial Operations, & GCF Factoring

Part A: Simplify Using Exponent Rules

Simplify each expression. Show your work.

Problem 1:

$$x^4 \cdot x^6$$

Problem 2:

$$\frac{y^9}{y^5}$$

Problem 3:

$$(m^3)^4$$

Problem 4:

$$(2x^2y^3)^3$$

Problem 5:

$$\frac{12x^7}{3x^2}$$

Part B: Polynomial Operations

Multiply the polynomials.

Problem 6:

$$3x(2x^2 - 5x + 4)$$

Problem 7:

$$(x + 4)(x + 7)$$

Problem 8:

$$(2x - 3)(x + 5)$$

Part C: GCF Factoring

Factor out the greatest common factor (GCF) from each expression.

Problem 9:

$$15y^2 - 10y$$

Problem 10:

$$8x^3 + 12x^2 - 4x$$

Problem 11:

$$14x^2y + 21xy^2$$

Part D: Word Problem

Problem 12:

The area of a square is given by $6x^2 + 9x$ square inches.

Part A:

Factor this expression.

Part B:

Find one possible combination for the length, width, and area of the square.

Hint: Use the definition of a square and your factored expression from Part A.

Session 9.2:

Factoring Trinomials & Special Products

Quick Reference: Factoring Trinomials

Factoring $x^2 + bx + c$

To factor a trinomial of the form $x^2 + bx + c$:

Steps:

1. Find two numbers that multiply to give c
2. AND add to give b
3. Write as $(x + \text{first number})(x + \text{second number})$

Factoring $ax^2 + bx + c$ (when $a \neq 1$)

AC Method:

1. Multiply $a \times c$
2. Find two numbers that multiply to ac and add to b
3. Rewrite the middle term using these two numbers
4. Factor by grouping

Special Products (Shortcuts)

Perfect Square Trinomials:

$$a^2 + 2ab + b^2 = (a + b)^2$$

$$a^2 - 2ab + b^2 = (a - b)^2$$

Difference of Squares:

$$a^2 - b^2 = (a + b)(a - b)$$

Homework 9.2:

Factoring Trinomials & Special Products

Part A: Factor Simple Trinomials ($x^2 + bx + c$)

Factor each trinomial completely.

Problem 1:

$$x^2 + 8x + 15$$

Problem 2:

$$x^2 - 7x + 12$$

Problem 3:

$$x^2 + 2x - 15$$

Part B: Factor When $a \neq 1$

Use the AC method to factor.

Problem 4:

$$2x^2 + 7x + 3$$

Problem 5:

$$3x^2 + 10x + 8$$

Problem 6:

$$2x^2 - 5x - 3$$

Part C: Perfect Square Trinomials

Determine if each is a perfect square. If yes, factor it.

Problem 7:

$$x^2 + 10x + 25$$

Problem 8:

$$4x^2 + 12x + 9$$

Part D: Difference of Squares

Factor each difference of squares.

Problem 9:

$$x^2 - 49$$

Problem 10:

$$25x^2 - 64$$

Part E: Mixed Practice

Factor completely. Check for GCF first!

Problem 11:

$$2x^2 - 50$$

Problem 12:

$$3x^2 + 18x + 27$$

Part F: Word Problem

Problem 13:

A square garden has an area of $(x^2 + 14x + 49)$ square feet.

Factor this expression to find the length of one side of the garden.

If $x = 6$ feet, what is the length of one side?

Session 9.3

Solve Quadratics by Factoring; Applications

Quick Reference: Solving by Factoring

Zero Product Property

If $a \cdot b = 0$, then $a = 0$ or $b = 0$ (or both).

This is the key to solving quadratic equations by factoring!

Steps to Solve by Factoring

1. Write the equation in standard form: $ax^2 + bx + c = 0$
2. Factor the left side completely
3. Set each factor equal to zero
4. Solve each equation
5. Check your solutions (optional but recommended)

Homework 9.3: Solving Quadratics by Factoring

Part A: Solve by Factoring

Solve each equation by factoring. Show all steps.

Problem 1:

$$x^2 + 7x + 10 = 0$$

Problem 2:

$$x^2 - 4x - 12 = 0$$

Problem 3:

$$x^2 - 64 = 0$$

Problem 4:

$$2x^2 + 9x + 4 = 0$$

Part B: Equations Not in Standard Form

Rewrite in standard form, then solve by factoring.

Problem 5:

$$x^2 + 8x = -15$$

Problem 6:

$$x^2 = 5x + 14$$

Part C: Equations with GCF

Factor out the GCF first, then solve.

Problem 7:

$$3x^2 = 27x$$

Part D: Application Problems

Problem 8: Area of a Rectangle

A rectangle has a length that is 4 meters more than its width.

The area of the rectangle is 60 meters squared.

Find the dimensions of the rectangle.

Let w = width in meters

Then length = _____ meters

Area equation: _____

Rewrite in standard form:

Factor and solve:

Which solution makes sense in context? _____

Width: _____ meters

Length: _____ meters

Problem 9: Projectile Motion

A ball is thrown upward from the ground.

Its height h in feet after t seconds is given by $h = -16t^2 + 48t$.

When will the ball hit the ground?

Hint: The ball hits the ground when $h = 0$.

Unit 9 Quiz: Quadratics & Polynomials

Section 1: Exponent Rules

Simplify each expression completely. Show your work.

Problem 1:

$$x^5 \cdot x^8$$

Problem 2:

$$(3xy^2)^3$$

Problem 3:

$$\frac{20m^9}{4m^3}$$

Section 2: Polynomial Operations

Multiply the polynomials. Show your work.

Problem 4:

$$5x(3x^2 - 2x + 7)$$

Problem 5:

$$(3x - 2)(2x + 5)$$

Section 3: GCF Factoring

Factor out the greatest common factor (GCF) from each expression.

Problem 6:

$$18x^3 - 12x^2$$

Problem 7:

$$15a^2b^3 + 25ab^2$$

Section 4: Factoring Trinomials & Special Products

Factor each expression completely.

Problem 8

$$x^2 + 11x + 24$$

Problem 9:

$$2x^2 + 11x + 5$$

Section 5: Special Products

Factor each expression completely.

Problem 10:

$$x^2 + 16x + 64$$

Problem 11:

$$x^2 - 81$$

Section 6: Solving by Factoring

Solve each equation by factoring. Show all steps.

Problem 12:

$$x^2 + 9x + 20 = 0$$

Problem 13:

$$2x^2 - 18 = 0$$

Section 7: Word Problems

Show all work, including your equation setup.

Problem 14:

A rectangular garden has a length that is twice its width.

The area of the garden is 98 square feet.

Find the dimensions of the garden.

Problem 15:

A rocket is launched upward from the ground.

Its height h in feet after t seconds is given by $h = -4t^2 + 32t$.

At what time is the rocket at a height of 64 feet?

Homework 9.1 Answer Key:

Exponents, Polynomial Operations, & GCF Factoring

Part A: Simplify Using Exponent Rules

Problem 1: $x^4 \cdot x^6$

Solution: Using the product rule, add the exponents.

$$x^4 \cdot x^6 = x^{4+6} = x^{10}$$

Answer: x^{10}

Problem 2: $\frac{y^9}{y^5}$

Solution: Using the quotient rule, subtract the exponents.

$$\frac{y^9}{y^5} = y^{9-5} = y^4$$

Answer: y^4

Problem 3: $(m^3)^4$

Solution: Using the power rule, multiply the exponents.

$$(m^3)^4 = m^{3 \cdot 4} = m^{12}$$

Answer: m^{12}

Problem 4: $(2x^2y^3)^3$

Solution: Apply the power to each factor.

$$(2x^2y^3)^3 = 2^3 \cdot (x^2)^3 \cdot (y^3)^3 = 8x^6y^9$$

Answer: $8x^6y^9$

Problem 5: $\frac{12x^7}{3x^2}$

Solution: Divide coefficients and subtract exponents.

$$\frac{12x^7}{3x^2} = \frac{12}{3} \cdot x^{7-2} = 4x^5$$

Answer: $4x^5$

Part B: Polynomial Operations

Problem 6: $3x(2x^2 - 5x + 4)$

Solution: Distribute $3x$ to each term.

$$3x(2x^2) + 3x(-5x) + 3x(4) = 6x^3 - 15x^2 + 12x$$

Answer: $6x^3 - 15x^2 + 12x$

Problem 7: $(x + 4)(x + 7)$

Solution: Use FOIL method.

First: $x \cdot x = x^2$

Outer: $x \cdot 7 = 7x$

Inner: $4 \cdot x = 4x$

Last: $4 \cdot 7 = 28$

Combine: $x^2 + 7x + 4x + 28 = x^2 + 11x + 28$

Answer: $x^2 + 11x + 28$

Problem 8: $(2x - 3)(x + 5)$

Solution: Use FOIL method.

First: $2x \cdot x = 2x^2$

Outer: $2x \cdot 5 = 10x$

Inner: $-3 \cdot x = -3x$

Last: $-3 \cdot 5 = -15$

Combine: $2x^2 + 10x - 3x - 15 = 2x^2 + 7x - 15$

Answer: $2x^2 + 7x - 15$

Part C: GCF Factoring

Problem 9: $15y^2 - 10y$

Solution: GCF is $5y$.

$$15y^2 - 10y = 5y(3y) + 5y(-2) = 5y(3y - 2)$$

Answer: $5y(3y - 2)$

Problem 10: $8x^3 + 12x^2 - 4x$

Solution: GCF is $4x$.

$$8x^3 + 12x^2 - 4x = 4x(2x^2) + 4x(3x) + 4x(-1) = 4x(2x^2 + 3x - 1)$$

Answer: $4x(2x^2 + 3x - 1)$

Problem 11: $14x^2y + 21xy^2$

Solution: GCF is $7xy$.

$$14x^2y + 21xy^2 = 7xy(2x) + 7xy(3y) = 7xy(2x + 3y)$$

Answer: $7xy(2x + 3y)$

Part D: Word Problem

Problem 12:

The area of a square is given by $6x^2 + 9x$ square inches.

Part A: Factor this expression.

Solution: GCF is $3x$.

$$6x^2 + 9x = 3x(2x + 3)$$

Answer: $3x(2x + 3)$

Part B: Find one possible combination for the length, width, and area of the square.

Solution: Note: The problem statement says "square" but the factored form gives different dimensions, suggesting this is actually a rectangle.

From the factored form $3x(2x + 3)$:

$$\text{Width} = 3x \text{ inches}$$

$$\text{Length} = (2x + 3) \text{ inches}$$

$$\text{Area} = 6x^2 + 9x \text{ square inches}$$

For a specific example, if $x = 1$:

$$\text{Width} = 3(1) = 3 \text{ inches}$$

$$\text{Length} = 2(1) + 3 = 5 \text{ inches}$$

$$\text{Area} = 6(1)^2 + 9(1) = 15 \text{ square inches}$$

Answer: Width = $3x$, Length = $(2x + 3)$; or if $x = 1$: Width = 3 in, Length = 5 in, Area = 15 sq in

Homework 9.2 Answer Key:

Factoring Trinomials & Special Products

Part A: Factor Simple Trinomials ($x^2 + bx + c$)

Problem 1: $x^2 + 8x + 15$

Solution: Find two numbers that multiply to 15 and add to 8.

Numbers: 3 and 5 (since $3 \times 5 = 15$ and $3 + 5 = 8$)

Answer: $(x + 3)(x + 5)$

Problem 2: $x^2 - 7x + 12$

Solution: Find two numbers that multiply to 12 and add to -7 .

Numbers: -3 and -4 (since $(-3) \times (-4) = 12$ and $-3 + (-4) = -7$)

Answer: $(x - 3)(x - 4)$

Problem 3: $x^2 + 2x - 15$

Solution: Find two numbers that multiply to -15 and add to 2.

Numbers: 5 and -3 (since $5 \times (-3) = -15$ and $5 + (-3) = 2$)

Answer: $(x + 5)(x - 3)$

Part B: Factor When $a \neq 1$

Problem 4: $2x^2 + 7x + 3$

Solution: Use AC method. $a \times c = 2 \times 3 = 6$

Find two numbers that multiply to 6 and add to 7: 6 and 1

$$2x^2 + 6x + 1x + 3$$

$$2x(x + 3) + 1(x + 3)$$

Answer: $(2x + 1)(x + 3)$

Problem 5: $3x^2 + 10x + 8$

Solution: Use AC method. $a \times c = 3 \times 8 = 24$

Find two numbers that multiply to 24 and add to 10: 6 and 4

$$3x^2 + 6x + 4x + 8$$

$$3x(x + 2) + 4(x + 2)$$

Answer: $(3x + 4)(x + 2)$

Problem 6: $2x^2 - 5x - 3$

Solution: Use AC method. $a \times c = 2 \times (-3) = -6$

Find two numbers that multiply to -6 and add to -5 : -6 and 1

$$2x^2 - 6x + 1x - 3$$

$$2x(x - 3) + 1(x - 3)$$

Answer: $(2x + 1)(x - 3)$

Part C: Perfect Square Trinomials

Problem 7: $x^2 + 10x + 25$

Solution: Check if it's a perfect square: $x^2 + 2(x)(5) + 5^2$

Yes! This is $(x + 5)^2$

Answer: $(x + 5)^2$

Problem 8: $4x^2 + 12x + 9$

Solution: Check if it's a perfect square: $(2x)^2 + 2(2x)(3) + 3^2$

Yes! This is $(2x + 3)^2$

Answer: $(2x + 3)^2$

Part D: Difference of Squares

Problem 9: $x^2 - 49$

Solution: This is $x^2 - 7^2$, a difference of squares.

Answer: $(x + 7)(x - 7)$

Problem 10: $25x^2 - 64$

Solution: This is $(5x)^2 - 8^2$, a difference of squares.

Answer: $(5x + 8)(5x - 8)$

Part E: Mixed Practice

Problem 11: $2x^2 - 50$

Solution: First factor out GCF of 2.

$$2(x^2 - 25)$$

Then recognize difference of squares: $x^2 - 5^2$

$$2(x + 5)(x - 5)$$

Answer: $2(x + 5)(x - 5)$

Problem 12: $3x^2 + 18x + 27$

Solution: First factor out GCF of 3.

$$3(x^2 + 6x + 9)$$

Then recognize perfect square trinomial: $x^2 + 2(x)(3) + 3^2$

$$3(x + 3)^2$$

Answer: $3(x + 3)^2$

Part F: Word Problem

Problem 13:

A square garden has an area of $(x^2 + 14x + 49)$ square feet.

Factor this expression to find the length of one side of the garden.

Solution: This is a perfect square trinomial: $x^2 + 2(x)(7) + 7^2$

$$(x + 7)^2$$

Since area = (side)² for a square, one side = $(x + 7)$ feet

Answer: $(x + 7)$ feet

If $x = 6$ feet, what is the length of one side?

Solution: Substitute $x = 6$: Side = $6 + 7 = 13$ feet

Answer: 13 feet

Homework 9.3 Answer Key: Solving Quadratics by Factoring

Part A: Solve by Factoring

Problem 1: $x^2 + 7x + 10 = 0$

Solution:

Factor: $(x + 2)(x + 5) = 0$

Set each factor equal to zero:

$$x + 2 = 0 \text{ or } x + 5 = 0$$

$$x = -2 \text{ or } x = -5$$

Answer: $x = -2$ or $x = -5$

Problem 2: $x^2 - 4x - 12 = 0$

Solution:

Factor: $(x - 6)(x + 2) = 0$

Set each factor equal to zero:

$$x - 6 = 0 \text{ or } x + 2 = 0$$

$$x = 6 \text{ or } x = -2$$

Answer: $x = 6$ or $x = -2$

Problem 3: $x^2 - 64 = 0$

Solution:

Factor (difference of squares): $(x + 8)(x - 8) = 0$

Set each factor equal to zero:

$$x + 8 = 0 \text{ or } x - 8 = 0$$

$$x = -8 \text{ or } x = 8$$

Answer: $x = -8$ or $x = 8$

Problem 4: $2x^2 + 9x + 4 = 0$

Solution:

Factor: $(2x + 1)(x + 4) = 0$

Set each factor equal to zero:

$$2x + 1 = 0 \text{ or } x + 4 = 0$$

$$x = -\frac{1}{2} \text{ or } x = -4$$

Answer: $x = -\frac{1}{2}$ or $x = -4$

Part B: Equations Not in Standard Form

Problem 5: $x^2 + 8x = -15$

Solution:

Rewrite in standard form: $x^2 + 8x + 15 = 0$

Factor: $(x + 3)(x + 5) = 0$

Set each factor equal to zero:

$$x + 3 = 0 \text{ or } x + 5 = 0$$

$$x = -3 \text{ or } x = -5$$

Answer: $x = -3$ or $x = -5$

Problem 6: $x^2 = 5x + 14$

Solution:

Rewrite in standard form: $x^2 - 5x - 14 = 0$

Factor: $(x - 7)(x + 2) = 0$

Set each factor equal to zero:

$$x - 7 = 0 \text{ or } x + 2 = 0$$

$$x = 7 \text{ or } x = -2$$

Answer: $x = 7$ or $x = -2$

Part C: Equations with GCF

Problem 7: $3x^2 = 27x$

Solution:

Rewrite in standard form: $3x^2 - 27x = 0$

Factor out GCF: $3x(x - 9) = 0$

Set each factor equal to zero:

$$3x = 0 \text{ or } x - 9 = 0$$

$$x = 0 \text{ or } x = 9$$

Answer: $x = 0$ or $x = 9$

Part D: Application Problems

Problem 8: Area of a Rectangle

A rectangle has a length that is 4 meters more than its width.

The area of the rectangle is 60 meters squared.

Find the dimensions of the rectangle.

Solution:

Let w = width in meters

Then length = $w + 4$ meters

Area equation: $w(w + 4) = 60$

Rewrite in standard form: $w^2 + 4w - 60 = 0$

Factor: $(w + 10)(w - 6) = 0$

Set each factor equal to zero:

$$w + 10 = 0 \text{ or } w - 6 = 0$$

$$w = -10 \text{ or } w = 6$$

Which solution makes sense in context? $w = 6$ (width cannot be negative)

Width: 6 meters

Length: $6 + 4 = 10$ meters

Answer: Width = 6 meters, Length = 10 meters

Problem 9: Projectile Motion

A ball is thrown upward from the ground.

Its height h in feet after t seconds is given by $h = -16t^2 + 48t$.

When will the ball hit the ground?

Solution:

The ball hits the ground when $h = 0$:

$$-16t^2 + 48t = 0$$

Factor out GCF: $-16t(t - 3) = 0$

Set each factor equal to zero:

$$-16t = 0 \text{ or } t - 3 = 0$$

$t = 0$ or $t = 3$

$t = 0$ is when the ball is thrown (starting position)

$t = 3$ is when the ball hits the ground

Answer: The ball hits the ground at $t = 3$ seconds

Unit 9 Quiz Answer Key: Quadratics & Polynomials

Section 1: Exponent Rules

Problem 1: $x^5 \cdot x^8$

Solution: Using the product rule, add the exponents.

$$x^5 \cdot x^8 = x^{5+8} = x^{13}$$

Answer: x^{13}

Problem 2: $(3xy^2)^3$

Solution: Apply the power to each factor.

$$(3xy^2)^3 = 3^3 \cdot x^3 \cdot (y^2)^3 = 27x^3y^6$$

Answer: $27x^3y^6$

Problem 3: $\frac{20m^9}{4m^3}$

Solution: Divide coefficients and subtract exponents.

$$\frac{20m^9}{4m^3} = \frac{20}{4} \cdot m^{9-3} = 5m^6$$

Answer: $5m^6$

Section 2: Polynomial Operations

Problem 4: $5x(3x^2 - 2x + 7)$

Solution: Distribute $5x$ to each term.

$$5x(3x^2) + 5x(-2x) + 5x(7) = 15x^3 - 10x^2 + 35x$$

Answer: $15x^3 - 10x^2 + 35x$

Problem 5: $(3x - 2)(2x + 5)$

Solution: Use FOIL method.

First: $3x \cdot 2x = 6x^2$

Outer: $3x \cdot 5 = 15x$

Inner: $-2 \cdot 2x = -4x$

Last: $-2 \cdot 5 = -10$

Combine: $6x^2 + 15x - 4x - 10 = 6x^2 + 11x - 10$

Answer: $6x^2 + 11x - 10$

Section 3: GCF Factoring

Problem 6: $18x^3 - 12x^2$

Solution: GCF is $6x^2$.

$$18x^3 - 12x^2 = 6x^2(3x) + 6x^2(-2) = 6x^2(3x - 2)$$

Answer: $6x^2(3x - 2)$

Problem 7: $15a^2b^3 + 25ab^2$

Solution: GCF is $5ab^2$.

$$15a^2b^3 + 25ab^2 = 5ab^2(3ab) + 5ab^2(5) = 5ab^2(3ab + 5)$$

Answer: $5ab^2(3ab + 5)$

Section 4: Factoring Trinomials & Special Products

Problem 8: $x^2 + 11x + 24$

Solution: Find two numbers that multiply to 24 and add to 11.

Numbers: 3 and 8 (since $3 \times 8 = 24$ and $3 + 8 = 11$)

Answer: $(x + 3)(x + 8)$

Problem 9: $2x^2 + 11x + 5$

Solution: Use AC method. $a \times c = 2 \times 5 = 10$

Find two numbers that multiply to 10 and add to 11: 10 and 1

$$2x^2 + 10x + 1x + 5$$

$$2x(x + 5) + 1(x + 5)$$

Answer: $(2x + 1)(x + 5)$

Section 5: Special Products

Problem 10: $x^2 + 16x + 64$

Solution: Recognize as a perfect square trinomial.

$$x^2 + 2(x)(8) + 8^2 = (x + 8)^2$$

Answer: $(x + 8)^2$

Problem 11: $x^2 - 81$

Solution: Recognize as difference of squares.

$$x^2 - 9^2 = (x + 9)(x - 9)$$

Answer: $(x + 9)(x - 9)$

Section 6: Solving by Factoring

Problem 12: $x^2 + 9x + 20 = 0$

Solution:

Factor: $(x + 4)(x + 5) = 0$

Set each factor equal to zero:

$$x + 4 = 0 \text{ or } x + 5 = 0$$

$$x = -4 \text{ or } x = -5$$

Answer: $x = -4$ or $x = -5$

Problem 13: $2x^2 - 18 = 0$

Solution:

Factor out GCF: $2(x^2 - 9) = 0$

Recognize difference of squares: $2(x + 3)(x - 3) = 0$

Set each factor equal to zero:

$$x + 3 = 0 \text{ or } x - 3 = 0$$

$$x = -3 \text{ or } x = 3$$

Answer: $x = -3$ or $x = 3$

Section 7: Word Problems

Problem 14:

A rectangular garden has a length that is twice its width.

The area of the garden is 98 square feet.

Find the dimensions of the garden.

Solution:

Let w = width in feet

Then length = $2w$ feet

Area equation: $w \cdot 2w = 98$

$$2w^2 = 98$$

$$w^2 = 49$$

$$w^2 - 49 = 0$$

$$\text{Factor: } (w + 7)(w - 7) = 0$$

$$w = -7 \text{ or } w = 7$$

Width must be positive, so $w = 7$ feet

$$\text{Length} = 2(7) = 14 \text{ feet}$$

Answer: Width = 7 feet, Length = 14 feet

Problem 15:

A rocket is launched upward from the ground.

Its height h in feet after t seconds is given by $h = -4t^2 + 32t$.

At what time is the rocket at a height of 64 feet?

Solution:

$$\text{Set } h = 64: 64 = -4t^2 + 32t$$

$$\text{Rewrite in standard form: } -4t^2 + 32t - 64 = 0$$

$$\text{Factor out GCF of } -4: -4(t^2 - 8t + 16) = 0$$

$$\text{Factor the trinomial (perfect square): } -4(t - 4)^2 = 0$$

$$\text{Set factor equal to zero: } (t - 4)^2 = 0$$

$$t - 4 = 0$$

$$t = 4$$

Answer: The rocket is at 64 feet at $t = 4$ seconds